MAT 017B Midterm 2

August 30, 2018

Name: Key

SID:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1 \qquad \int \frac{1}{x}\,dx = \ln|x| + C \qquad \int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C \qquad \int \sec^2 x\,dx = \tan x + C$$

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C \qquad \int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int_a^b f(x)\,dx \approx \sum_{k=1}^{n} f\left(x_k\right)\Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x$$

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x) \qquad \frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt = f[g(x)]g'(x) - f[h(x)]h'(x)$$

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C \qquad \int u\,dv = uv - \int v\,du$$

$$A = \int [f(x) - g(x)]\,dx \quad \Delta y = \int \frac{dy}{dx}\,dx \quad \bar{y} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$f(x) \approx \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \quad \frac{dy}{dx} = f(x)g(y) \quad \int \frac{dy}{g(y)} = \int f(x)\,dx$$

$$y' + p(x)y = q(x) \quad I(x) = e^{\int p(x)dx} \quad y = \frac{1}{I(x)}\int I(x)q(x)\,dx$$

1. (10 points) A tree has a height of 50 meters. Find the total biomass in kilograms of the tree if the density of biomass in kilograms per meter along the height of a tree is given by the equation

$$\frac{dM}{dh} = 30 + \frac{1}{10}h + \frac{1}{10}e^{\frac{1}{10}h}.$$

$$\int_0^M dM = \int_0^{50} 30 + \frac{1}{10}h + \frac{1}{10}e^{\frac{1}{10}h}\, dh$$

$$M = 30h + \frac{1}{20}h^2 + e^{\frac{1}{10}h} \Big|_{h=0}^{h=50}$$

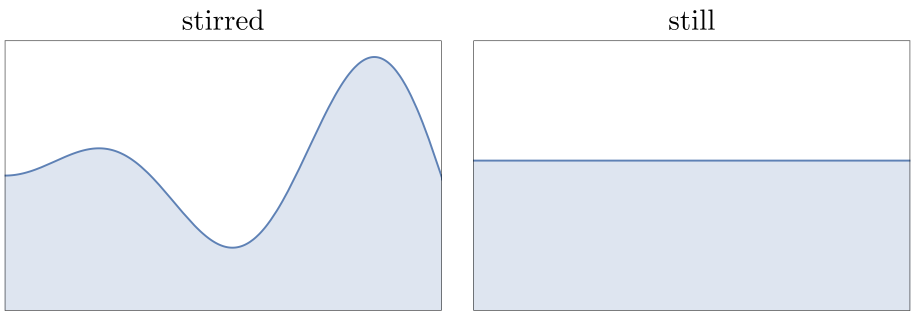
$$= 30(50) + \frac{1}{20}(50)^2 + e^{\frac{50}{10}} - 30(0) - \frac{1}{20}(0)^2 - e^{\frac{0}{10}}$$

$$= 1500 + \frac{2500}{20} + e^5 - 1$$

$$= 1500 + 125 + e^5 - 1$$

$$\boxed{M = 1624 + e^5 \text{ kg}}$$

2. (10 points) The height in inches of the water in a tank when stirred is observed to follow the equation $h = x \sin x + 9$ starting from $x = 0$ in. to $x = 3\pi$ in. What will the height in inches of the water be once the water is still? (i.e. Find the average height of the water.)



$$\bar{h} = \frac{1}{3\pi - 0}\int_0^{3\pi} x \sin x + 9 \, dx$$

$$u = x \qquad dv = \sin x \, dx$$

$$du = dx \qquad v = -\cos x$$

$$= \frac{1}{3\pi}\left[-x\cos x + \int \cos x \, dx + 9x\right]_0^{3\pi}$$

$$= \frac{1}{3\pi}\left[-x\cos x + \sin x + 9x\right]_0^{3\pi}$$

$$= \frac{1}{3\pi}\left[-3\pi\cos 3\pi + \sin 3\pi + 9(3\pi) + 0\cos 0 - \sin 0 + 9(0)\right]$$

$$= \frac{1}{3\pi}\left[3\pi + 27\pi\right]$$

$$= \frac{30\pi}{3\pi}$$

$$\boxed{\bar{h} = 10 \text{ in.}}$$

3. (20 points) Find a Taylor polynomial of degree 4 of xe^{-x} about $x = 0$.

$$f'(x) = -xe^{-x} + e^{-x}$$

$$f''(x) = -(-xe^{-x} + e^{-x}) - e^{-x}$$

$$= xe^{-x} - e^{-x} - e^{-x}$$

$$= xe^{-x} - 2e^{-x}$$

$$f'''(x) = -xe^{-x} + e^{-x} + 2e^{-x}$$

$$= -xe^{-x} + 3e^{-x}$$

$$f^{(4)}(x) = -(-xe^{-x} + e^{-x}) - 3e^{-x}$$

$$= xe^{-x} - 4e^{-x}$$

$$f(0) = 0e^{-0} = 0$$

$$f'(0) = 0e^{-0} + e^{-0} = 1$$

$$f''(0) = 0e^{-0} - 2e^{-0} = -2$$

$$f'''(0) = -0e^{-0} + 3e^{-0} = 3$$

$$f^{(4)}(0) = 0e^{-0} - 4e^{-0} = -4$$

$$P_4(x) = 0 + \frac{1}{1!}(x-0) - \frac{2}{2!}(x-0)^2 + \frac{3}{3!}(x-0)^3 - \frac{4}{4!}(x-0)^4$$

$$\boxed{P_4(x) = x - x^2 + \frac{1}{2}x^3 - \frac{1}{6}x^4}$$

4. (25 points) Let S represent the amount in grams of nutrients in a lake at time t days. An incoming stream containing 5 g of nutrients per liter flows into the lake at a rate of 5 L/day. and the well-stirred mixture flows out of the lake at a rate of 10 L/day. The lake initially holds 1000 L.

a. (5 points) Show that the system can be modeled by the following differential equation

$$\frac{dN}{dt} = 25 - \frac{2}{200-t}N.$$

$$V = 1000 + 5t - 10t$$
$$= 1000 - 5t$$

$$\frac{dN}{dt} = \begin{bmatrix}\text{rate}\\ \text{in}\end{bmatrix} - \begin{bmatrix}\text{rate}\\ \text{out}\end{bmatrix}$$

$$= \left(\frac{5\,g}{L}\right)\left(\frac{5\,L}{day}\right) - \left(\frac{N}{1000-5t}\frac{g}{L}\right)\left(\frac{10\,L}{day}\right)$$

$$= 25 - \frac{10N}{1000-5t}$$

$$\boxed{\frac{dN}{dt} = 25 - \frac{2}{200-t}N}$$

b. (20 points) Solve the differential equation to find an equation for the amount of nutrients in the lake at time t given that the lake initially contains 500 g of nutrients.

$$\frac{dN}{dt} + \frac{2}{200-t}N = 25 \qquad p(t) = \frac{2}{200-t}$$

$$q(t) = 25$$

$$\int p(t)\,dt = \int \frac{2}{200-t}\,dt$$

$$= -2\ln|200-t|$$

$$e^{\int p(t)dt} = e^{-2\ln|200-t|} = e^{\ln|200-t|^{-2}} = (200-t)^{-2}$$

$$\frac{1}{(200-t)^2}\frac{dN}{dt} + \frac{1}{(200-t)^2}\frac{2}{200-t}N = \frac{25}{(200-t)^2}$$

$$\frac{d}{dt}\left[\frac{1}{(200-t)^2}N(t)\right] = \frac{25}{(200-t)^2}$$

$$\frac{1}{(200-t)^2}N(t) = \int \frac{25}{(200-t)^2}\,dt$$

$$= \frac{25}{200-t} + C$$

$$N(t) = 25(200-t) + C(200-t)^2$$

$$N(0) = 25(200-0) + C(200-0)^2$$

$$500 = 5000 + 40000C$$

$$C = \frac{-4500}{40000}$$

$$\boxed{N(t) = 25(200-t) - \frac{4500}{40000}(200-t)^2}$$

5. (25 points) Suppose r and K are positive constants. Then the Gompertz Growth Model is given by the following equation

$$\frac{dN}{dt} = -rN \ln \frac{N}{K}$$

a. (10 points) Show that $\hat{N} = K$ is an equilibrium point and classify its stability.

$$g(N) = -rN \ln \frac{N}{K}$$

$$g(K) = -rN \ln \frac{K}{K}$$

$$= -rN \ln 1$$

$$= -rN \cdot 0$$

$$g(K) = 0$$

$$g'(N) = -rN \cdot \frac{K}{N} \cdot \frac{1}{K} - r \ln \frac{N}{K}$$

$$= -r - r \ln \frac{N}{K}$$

$$g'(K) = -r - r \ln \frac{K}{K}$$

$$= -r - r \ln \frac{K}{K}$$

$$= -r - r \cdot 0$$

$$= -r < 0 \Rightarrow \boxed{K \text{ is a stable equilibrium point.}}$$

b. (15 points) Solve the differential equation for $N(t)$ with the initial condition $N(0) = N_0$.

$$\frac{dN}{dt} = -rN\ln\frac{N}{K}$$

$$\int \frac{dN}{N\ln\frac{N}{K}} = \int -r\,dt$$

$$u = \ln\frac{N}{K}$$

$$du = \frac{K}{N}\cdot\frac{1}{K}dN$$

$$\int \frac{1}{u}du = -rt + C$$

$$\ln\left|\ln\frac{N}{K}\right| = -rt + C$$

$$\ln\frac{N}{K} = Ce^{-rt}$$

$$\frac{N}{K} = e^{Ce^{-rt}}$$

$$N = Ke^{Ce^{-rt}}$$

$$N(0) = Ke^{Ce^{-r\cdot 0}}$$

$$= Ke^{C} = N_0$$

$$\frac{N_0}{K} = e^{C}$$

$$\ln\frac{N_0}{K} = C$$

$$\boxed{N(t) = Ke^{\ln\left(\frac{N_0}{K}\right)e^{-rt}}}$$